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(%i4) /* Symbols */
declare(x, real)$;
declare([h, x_k], [real, constant])$;
assume(h > 0)$;
depends(f, x)$;
```

```
(%i6) /* Integrals */
I_simps: h / 3 * (f(x_k - h) + 4 * f(x_k) + f(x_k + h)); /* General triplet of points */
I_exact: 'integrate(f(x), x, x_k - h, x_k + h); /* Exact integral of the function in the triplet's interval */
```

$$I_simps \quad \frac{h \left(f(x_k + h) + f(x_k - h) + 4 f(x_k) \right)}{3}$$

$$I_exact \quad \int_{x_k - h}^{x_k + h} f(x) dx$$

```
(%i8) /* Expand the function in Taylor series */
subst(f(x) = Taylor(f(x), x, x_k, 4), I_exact)$;
I_exact: ev(%, nouns); /* I_exact approximated by Taylor series */
```

$$I_exact \quad \frac{h^5 \left(\frac{d^4}{dx^4} f(x) \right) \Big|_{x=x_k} + 20 h^3 \left(\frac{d^2}{dx^2} f(x) \right) \Big|_{x=x_k} + 120 h f(x_k)}{60}$$

```
(%i11) /* Expand f(x_k +/- h) in Taylor series */
f_minus_h: f(x_k)
```

$$\begin{aligned} & - \text{at}(\text{diff}(f(x), x, 1), x = x_k) \cdot h \\ & + \text{at}(\text{diff}(f(x), x, 2), x = x_k) \cdot h^2 / 2 \\ & - \text{at}(\text{diff}(f(x), x, 3), x = x_k) \cdot h^3 / 6 \\ & + \text{at}(\text{diff}(f(x), x, 4), x = x_k) \cdot h^4 / 24; \end{aligned}$$

f_plus_h: f(x_k)

$$\begin{aligned} & + \text{at}(\text{diff}(f(x), x, 1), x = x_k) \cdot h \\ & + \text{at}(\text{diff}(f(x), x, 2), x = x_k) \cdot h^2 / 2 \\ & + \text{at}(\text{diff}(f(x), x, 3), x = x_k) \cdot h^3 / 6 \\ & + \text{at}(\text{diff}(f(x), x, 4), x = x_k) \cdot h^4 / 24; \end{aligned}$$

```
I_simps: subst([ /* I_simps approximated by Taylor series */
f(x_k - h) = f_minus_h,
f(x_k + h) = f_plus_h
], I_simps);
```

$$I_simps \quad \frac{h \left(\frac{h^4 \left(\frac{d^4}{dx^4} f(x) \right) \Big|_{x=x_k}}{12} + h^2 \left(\frac{d^2}{dx^2} f(x) \right) \Big|_{x=x_k} + 6 f(x_k) \right)}{3}$$

```
(%i12) err: abs(expand(I_exact - I_simps)); /* Absolute error definition for a triplet */
```

err

$$\frac{h^5}{180} \left| \frac{d^4}{dx^4} f(x) \right|_{x=x_k}$$

(%i13) **total_err_h:** err · (b – a) / (2 · h); /* Total error full integral */

total_err_h

$$\frac{h^4 (b-a)}{180} \left| \frac{d^4}{dx^4} f(x) \right|_{x=x_k}$$

(%i14) **total_err_N:** subst(h = (b – a) / N, total_err_h); /* Alternative formula with number of points */

total_err_N

$$\frac{(b-a)^5}{180 N^4} \left| \frac{d^4}{dx^4} f(x) \right|_{x=x_k}$$